



PROJECT TITLE

DataQuest 2026 - Building Interpretable Credit Models



DATA DRIVEN

Harnessing the power of data to uncover insights and drive decisions.



RISK AWARE

Identifying and managing risk to build more resilient models.



INTERPRETABLE

Building transparent models that are explainable and trustworthy.



IMPACT FOCUSED

Delivering actionable insights that create measurable value.



PREPARED BY

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1. WHAT IS CREDIT RISK?

Credit risk is the risk a bank faces that a borrower may fail to meet their contractual repayment obligations.



Since lending is a core activity of banks, borrower defaults can lead to financial losses, reduced profitability, and lower capital adequacy.



To better understand and manage this risk, banks quantify it using statistical and financial techniques.

This process is known as **credit risk modelling**, and it helps banks make decisions such as whether to approve or reject loan applications.

3 KEY METRICS IN CREDIT RISK MODELLING



PD

Probability of Default

The likelihood that a borrower will fail to meet their contractual repayment obligations (e.g., missing payments for 90 days).



EAD

Exposure at Default

The total value the bank is exposed to at the time of default, including outstanding principal, accrued interest, and applicable fees.



LGD

Loss Given Default

The percentage of the Exposure at Default that the bank cannot recover after collateral, guarantees, or recoveries.

2. EXPECTED CREDIT LOSS (ECL)



These three metrics come together to estimate the expected loss from a loan due to borrower default.

$$\text{ECL} = \text{PD} \times \text{EAD} \times \text{LGD}$$



PD

Probability of Default

The likelihood that the borrower will default (e.g., miss payments for 90 days).



EAD

Exposure at Default

The total amount the bank is exposed to at the time of default (principal, interest, fees).



LGD

Loss Given Default

The percentage of EAD the bank cannot recover after collateral, guarantees, or recoveries.



ECL represents the expected loss a bank may incur from a loan due to borrower default.

We focus on modelling Probability of Default (PD).



Supervised Learning Models

PD models are typically supervised learning models trained on historical data where outcomes (default / no default) are already known.



Classification Problem

The objective is to classify borrowers according to risk. An application is classified as either a good borrower (0 = No Default) or a bad borrower (1 = Default).



From Probability to Classification

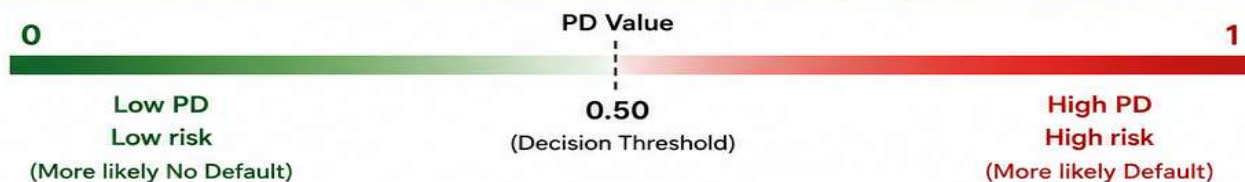
Models output a probability of default (PD) between 0 and 1. A threshold is used to convert probabilities into classes. A commonly used default threshold is 0.50.

Classification Rule (Threshold = 0.50)

$$\text{Loan/Application} = \begin{cases} \text{Rejected / Default,} & \text{if PD} \geq 0.50 \\ \text{Approved / No Default,} & \text{if PD} < 0.50 \end{cases}$$

Interpretation

The higher the PD, the higher the risk of default.



Logistic Regression (Linear Model)

Logistic Regression is a member of the **Generalized Linear Models (GLM)** family, just like Linear Regression.

Linear Regression	Logistic Regression
 Outcome: Continuous (e.g., amount, income)	 Outcome: Categorical (Typically binary: 0 or 1)
 Predicts any real value $(-\infty, +\infty)$	 Predicts probability of event p where $0 \leq p \leq 1$

1 Why we don't model p directly?

If we try a linear model:

$$p = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

It can produce any value in $(-\infty, +\infty)$, which is not suitable for probability $[0, 1]$.

2 Model the odds instead

Odds of the event occurring:

$$\text{odds} = \frac{p}{1 - p}$$

- Odds range: 0 to $+\infty$
- Not symmetric
- Cannot produce negative values

3 Take log of odds (logit function)

Logit (log-odds):

$$\text{logit}(p) = \ln\left(\frac{p}{1 - p}\right)$$

Maps probabilities from $(0, 1)$ to the entire real line $(-\infty, +\infty)$.

4 Assume linear relationship with features

Now we can model log-odds linearly:

$$\ln\left(\frac{p}{1 - p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where $x_1, x_2, \dots, x_n$ are features and $\beta_0, \beta_1, \dots, \beta_n$ are coefficients.

5 Convert back to probability (Sigmoid function)

Solve for p to get:

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

This is the sigmoid (logistic) function:

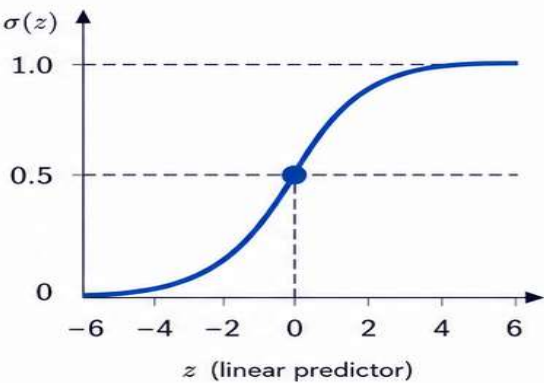
$$\sigma(z) = \frac{1}{1 + e^{-z}}, \text{ where } z = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

It ensures $0 \leq p \leq 1$ for any inputs.

The Sigmoid Function

The sigmoid function maps any real number z to a probability between 0 and 1.

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



Key Takeaway

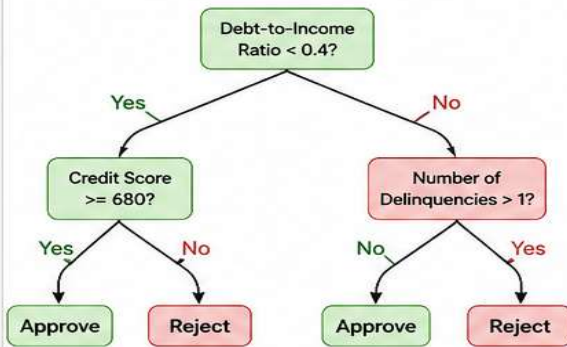
We use the sigmoid function to transform the linear predictor into a valid probability (0 to 1).

The higher the predicted probability (PD), the higher the risk of default.

Trees for Credit Risk Modeling (Approve/Reject)

1. Decision Tree

This model consists of a single tree used to make predictions.

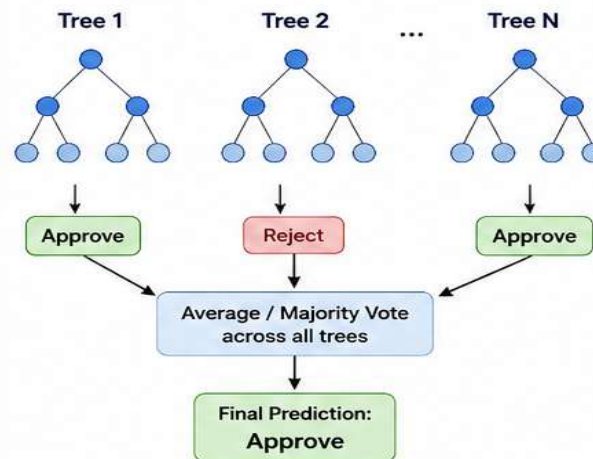


A single tree splits the data based on feature conditions and outputs a final prediction: Approve or Reject.

Prediction: Approve / Reject

2. Random Forest

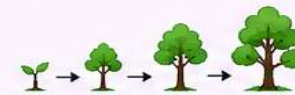
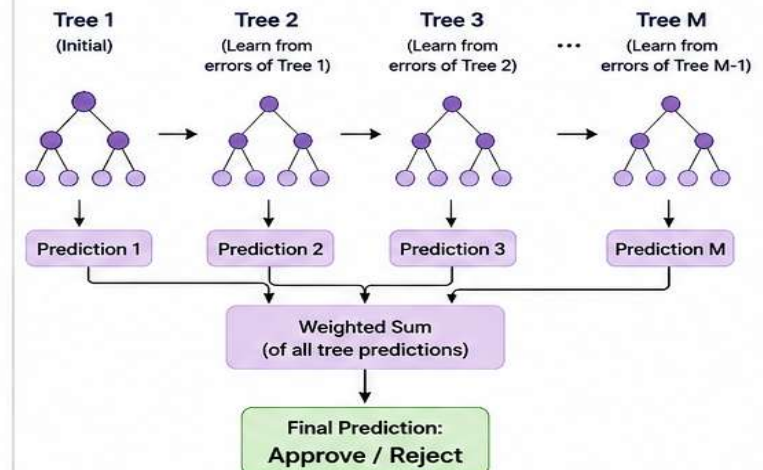
This model consists of many decision trees used for prediction, where the final prediction is typically obtained by averaging the predictions from all the trees.



Many trees built on different random samples and features. Their predictions are combined (averaged or voted) to produce the final result.

3. Gradient Boosting Machine

This model also uses many trees; however, each new tree is built to learn from and correct the mistakes made by the previous trees, improving the overall predictive performance.



Each new tree focuses on the residual errors of the previous ensemble, leading to progressively better predictions.



Goal:

Predict whether a loan application should be Approved or Rejected based on applicant features (e.g., credit score, income, debt-to-income ratio, delinquencies, employment history, etc.).

Approve (Low Risk)
Reject (High Risk)

Neural Networks (Non-Linear)



A neural network is a machine learning model that stacks simple computational units called neurons in multiple layers. These neurons learn patterns in the data by estimating weights and biases that map inputs to outputs.



At a high level, neural networks are inspired by biological neurons in the human brain, which communicate through electrical signals.

MATHEMATICAL REPRESENTATION

The structure of an artificial feed-forward neural network can be represented by the recursive expression:

$$\mathbf{a}^{(l)} = \sigma_l \left(\mathbf{W}_l^T \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)} \right), \quad l = 1, 2, \dots, L$$

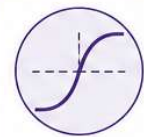
where:

- $\mathbf{a}^{(l)}$: vector of activation values at layer l
- $\mathbf{W}_l$: weight matrix for layer l
- $\mathbf{b}^{(l)}$: bias vector for layer l
- σ_l : activation function applied at layer l

The initial layer, $\mathbf{a}^{(0)}$, represents the **input features** for a particular observation, usually denoted by:

$$\mathbf{a}^{(0)} = \mathbf{x}_i$$

ACTIVATION FUNCTION AT OUTPUT LAYER

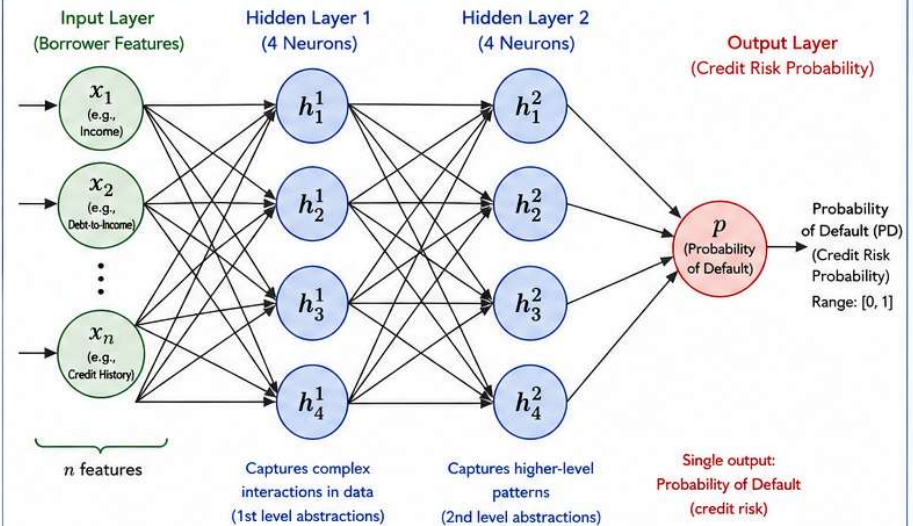


For binary classification problems (e.g., default vs. no default), the activation function applied at the final layer (L) is typically the **sigmoid function**.

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

This maps the output to a value between 0 and 1, allowing it to be interpreted as a probability of default.

ARTIFICIAL FEED-FORWARD NEURAL NETWORK



SIGMOID (LOGISTIC) ACTIVATION FUNCTION

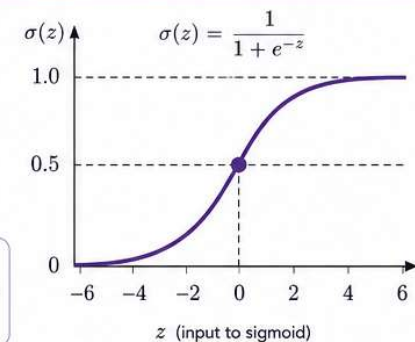
The sigmoid function maps any real-valued input z to a value between 0 and 1.

- Output range: (0, 1)
- Interpretable as probability
- Widely used in the output layer for binary classification



Interpretation:

Higher output $p \rightarrow$ Higher credit risk (more likely default)
Lower output $p \rightarrow$ Lower credit risk (less likely default)



Complexity and Interpretability



There is a trade-off between model **complexity** (ability to capture complex patterns) and **interpretability** (ease of understanding how predictions are made).

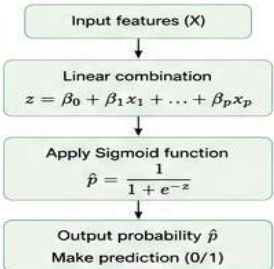


1. Simple Models: High Interpretability, Lower Complexity

Logistic Regression and Decision Trees are simple models. They have straightforward processes that are easy to follow. Therefore, they are **interpretable**, but may **struggle to capture** very complex structures in the data.

A. Logistic Regression

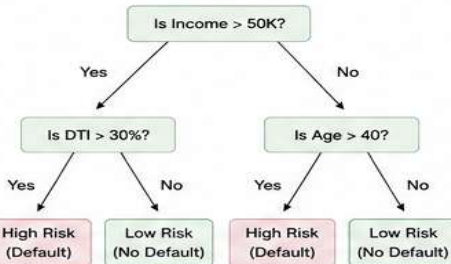
Simple & Direct Process



- Coefficients (β) show direction and strength of each feature.
- Easy to explain: changes in features lead to predictable changes in log-odds.

B. Decision Trees

Simple & Transparent Structure



- Easy to follow decision rules (if-then).
- We can trace the exact path for any prediction.



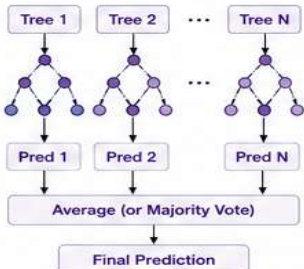
High Interpretability | **Low to Moderate Complexity**
Good for understanding and explaining predictions.

2. Complex Models: Low Interpretability, High Complexity

Random Forests, Gradient Boosting Machines (GBMs), and Neural Networks involve many interconnected steps and transformations before a prediction is made. It is easy to lose track of how predictions are produced, making them **less interpretable**, but they are **strong at capturing complex patterns**.

A. Random Forests

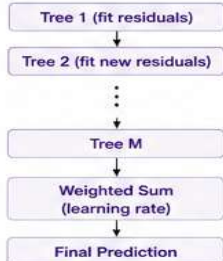
Many Trees → Aggregate



- Many trees built on different samples and features.
- Harder to interpret overall because of the ensemble.

B. Gradient Boosting Machines (GBMs)

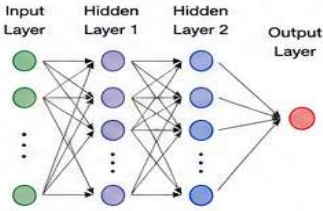
Sequential Trees → Correct Errors



- Trees are built sequentially, each trying to fix previous errors.
- Many steps and interactions are hard to interpret.

C. Neural Networks

Many Layers → Non-linear Transformations



- Many layers and weights.
- Highly non-linear transformations.
- Very powerful but acts like a "black box".



Low Interpretability | **High Complexity**
Powerful in capturing complex patterns, but harder to explain.

Summary Comparison

Model Type	Examples	Interpretability	Complexity	Strength
Simple Models	Logistic Regression, Decision Trees	★★★★★ High	★★☆☆☆ Low to Moderate	Easy to understand, transparent, but may miss complex patterns.
Complex Models	Random Forests, GBMs, Neural Networks	★☆☆☆☆ Low	★★★★★ High	Very powerful, capture complex structures, but harder to interpret.

Key Takeaways



There is a **trade-off**: as model complexity increases, **interpretability decreases**.



Simple models are preferred when **interpretability**, **transparency**, and **regulatory explainability** are important (e.g., credit scoring, risk models).



Complex models are preferred when **prediction accuracy** is the top priority and interpretability is less critical.

More Interpretable

Interpretability (High → Low)

More Complex

Complexity (Low → High)



Weights of Evidence (WoE) and Information Value (IV)

WoE measures the strength of a feature in separating good (non-events) and bad (events) loan applications.
IV measures the overall predictive power of the feature.



1. Weight of Evidence (WoE)

It is a measure of the strength of a feature (independent variable) in separating non-events and events. In our case, it measures how well a feature can separate good and bad loan applications.

$$WoE_i = \ln \left(\frac{\% \text{ of non-events}_i}{\% \text{ of events}_i} \right) = \ln \left(\frac{\text{Distribution of Good}_i}{\text{Distribution of Bad}_i} \right)$$

i refers to the i^{th} bin/category.



2. Information Value (IV)

It is used to measure the predictive power of independent variables. It helps us understand which independent variables provide the most useful information about the target variable.

$$IV = \sum_{i=1}^n (WoE_i \times (\% \text{ of non-events}_i - \% \text{ of events}_i))$$

IV Interpretation Guide	
IV Value	Predictive Power
$IV < 0.02$	Not predictive
$0.02 \leq IV < 0.1$	Weak
$0.1 \leq IV < 0.3$	Medium
$0.3 \leq IV < 0.5$	Strong
$IV \geq 0.5$	Suspiciously strong (possible overfitting or data leakage)

Key Takeaway



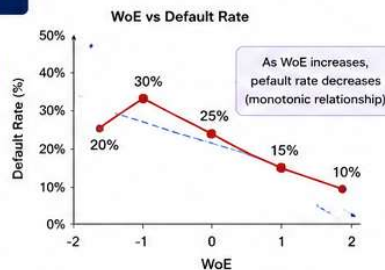
Higher IV → Stronger predictive power.

IV helps in variable selection: keep variables with medium to strong IV (typically $IV \geq 0.1$) and remove weak ones.

4. WoE Transforms Categories to Log-Odds Linearly

Example: Credit Utilization (%)

Utilization (Binned)	Good (Non-events)	Bad (Events)	% of Non-events	% of Events	WoE
0% - 20%	18,000	2,000	36.0%	10.0%	$\ln(0.36/0.10) = 1.28$
20% - 40%	16,000	3,000	32.0%	15.0%	$\ln(0.32/0.15) = 0.76$
40% - 60%	10,000	5,000	20.0%	25.0%	$\ln(0.20/0.25) = -0.22$
60% - 80%	4,000	6,000	8.0%	30.0%	$\ln(0.08/0.30) = -1.32$
80% - 100%	2,000	4,000	4.0%	20.0%	$\ln(0.04/0.20) = -1.61$



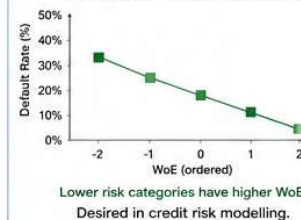
The plot shows a monotonic decreasing relationship:
Higher WoE → Lower default rate → Lower risk (better borrowers)

3. Why Weight of Evidence (WoE) is Important in Credit Modelling

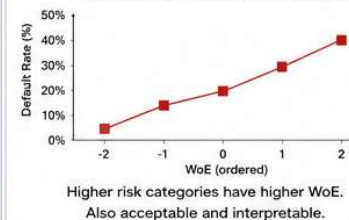
- ✓ It is typically used in logistic regression in credit modelling, which assumes a **linear relationship** between the **log-odds** of the target variable and the input features.
- ✓ WoE transforms **categorical** (or **binned continuous**) variables into numerical values that are **linearly related to the log-odds** of the target.
- ✓ This transformation makes the features more suitable for logistic regression and improves both **model stability** and **interpretability**.
- ✓ WoE groups categories that have similar distributions with respect to the target variable.
- ✓ This allows us to rank risk across categories by observing how the default rate changes from one group to another.

5. Ideal Relationship: Monotonic

Monotonic Decreasing (Good)
WoE increases → Default rate decreases



Monotonic Increasing (Also Good)
WoE increases → Default rate increases



We want the relationship to be **monotonic** (either consistently increasing or consistently decreasing) across ordered categories so that risk moves in one direction. This makes the model more interpretable and aligned with credit risk intuition.

Model Evaluation Metrics



We are modeling loan default (bad loan application) as the **positive class** (event) and non-default (good loan application) as the **negative class** (non-event).

1. Confusion Matrix

- **True Positive (TP):**
Actual default, predicted default.
- **False Negative (FN):**
Actual default, predicted non-default.
- **False Positive (FP):**
Actual non-default, predicted default.
- **True Negative (TN):**
Actual non-default, predicted non-default.

		Predicted	
		Predicted Positive (Default)	Predicted Negative (Non-default)
Actual	Actual Positive (Default)	TP	FN
	Actual Negative (Non-default)	FP	TN



The confusion matrix is the foundation for many classification evaluation metrics.

3 Recall (Sensitivity / True Positive Rate)

Out of all actual defaults (positive), how many did the model correctly identify.

$$\text{Recall} = \frac{TP}{TP + FN}$$



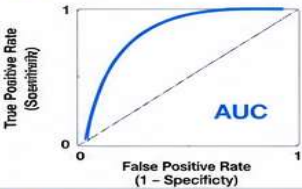
Focuses on capturing as many actual defaults as possible. High recall → few missed defaults (FN).

Actual Positive (Default)



Range: 0 to 1
(Higher is better)

5 AUC (Area Under the ROC Curve)



The Area Under the Receiver Operating Characteristic (ROC) curve. Measures how well the model separates defaults (events) from non-defaults (non-events) across all possible classification thresholds.

- AUC = 1.0 → Perfect separation
- AUC = 0.5 → No discrimination (random guessing)
- AUC < 0.5 → Worse than random

Range: 0 to 1
(Higher is better)



Key Takeaways

- Accuracy is useful but can be misleading with imbalanced data.
- Precision controls false positives (FP).
- Recall controls false negatives (FN).
- F1 Score balances precision and recall.
- AUC and Gini evaluate the model's ability to rank and separate classes across all thresholds.



When to Use What?

Metric	Best Used When...
Accuracy	Classes are balanced and overall correctness matters.
Precision	False positives are costly (e.g., unnecessary manual review, customer friction).
Recall	False negatives are costly (e.g., missing a potential default).
F1 Score	Need a balance between precision and recall.
AUC / Gini	Need to evaluate overall ranking power across all thresholds.



Choose metrics based on the business cost of FP vs. FN.

1 Accuracy (Overall Correctness)

The overall percentage of correct predictions (both default and non-default).

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$



Good for overall performance, especially when classes are balanced.



Range: 0 to 1
(Higher is better)

2 Precision (Positive Predictive Value)

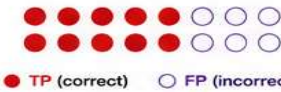
Out of all loan applications predicted to default (positive), how many actually defaulted.

$$\text{Precision} = \frac{TP}{TP + FP}$$



Focuses on the quality of positive predictions. High precision → few false alarms (FP).

Predicted Positive (Default)



Range: 0 to 1
(Higher is better)

4 F1 Score (Harmonic Mean of Precision & Recall)

Balances Precision and Recall. Useful when there is a trade-off between the two.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$



Higher F1 means a better balance between precision and recall.



Precision Recall

Range: 0 to 1
(Higher is better)

6 Gini Coefficient

Also measures how well the model separates defaults from non-defaults. It is directly related to AUC.

$$\text{Gini} = 2 \times \text{AUC} - 1$$

- Gini = 1 → Perfect model (AUC = 1)
- Gini = 0 → No discrimination (AUC = 0.5)
- Gini < 0 → Worse than random (AUC < 0.5)



Range: -1 to 1
(Higher is better)

Forbidden Features



 The features that a regulator will disapprove of are primarily those that **introduce discrimination** and those that **lack business interpretability**.

1. What Are Forbidden Features?



Forbidden features include **race, religion, gender, marital status**, etc.



They also include **proxy variables** of such attributes—features that are indirectly correlated with protected characteristics, such as zip code, geographical data, email domain, etc.



These features are disallowed because of **fair lending practices**. Models must not perpetuate bias or discrimination, and they must also support **explainability** and **accountability**.

2. Why Are They Forbidden?



Prevent Discrimination
Using protected attributes or their proxies can lead to unfair treatment of certain groups, violating equal opportunity and fair lending laws.



Comply with Regulations
Regulations such as the Equal Credit Opportunity Act (ECOA), FCRA, and GDPR prohibit discrimination based on protected traits.



Ensure Explainability & Accountability
Lenders must provide clear, business-justifiable reasons for adverse actions (e.g., loan rejection). Forbidden features hinder interpretability.



Maintain Trust & Fairness
Fair and transparent models build customer trust and support ethical, responsible lending.

3. Examples of Forbidden Features			
A. Directly Protected Attributes		B. Proxy (Indirect) Variables	
Explicit characteristics protected by law.		Features that may be correlated with protected attributes.	
 Race / Ethnicity	 Pregnancy / Parental Status	 Zip Code / Postal Code	 School / University
 Religion	 Age (in some jurisdictions)	 Geographical Data (e.g., neighborhood, census tract)	 Membership in Clubs / Organizations
 Gender / Sex	 Disability	 Email Domain (e.g., @freemail.com)	 Purchase History (certain categories)
 Marital Status	 Veteran Status	 Address / Street Information	 Device / IP / Browser Data (in some contexts)

4. Key Principles



Fair Lending
Do not use features that lead to disparate impact or unfair advantage.



Business Interpretability
Features must have a clear, legitimate business reason and be understandable.



Transparency
Models should support clear explanations for decisions.



Accountability
Lenders are responsible for outcomes and must justify decisions.

 **5. Bottom Line**

Avoid using protected attributes and their proxies. Build models that are fair, interpretable, and compliant—so every decision is justified, explainable, and free from bias.



1 Data Cleaning Approach

The dataset (loan book) contains **602 duplicate records**, which will be removed.



1

Remove Duplicates

The dataset (loan book) contains **602 duplicate records**, which will be removed.

2

Identify Missing Values

There are 4 variables with missing values:

- **annual_income:** 8638
- **employment_length_years:** 3711
- **num_open_accounts:** 2423
- **months_since_last_delinquency:** 60068

3

Drop High-Missingness Variable

The variable **months_since_last_delinquency** was dropped because nearly half of its observations contain missing values.

4

Impute annual_income by Region Median

For **annual_income**, the data was grouped by region, and the missing values were imputed using the median annual income within each region. The reason for this approach is that income levels may differ across locations.

5

Impute Other Numerical Variables by Median

For **employment_length_years** and **num_open_accounts**, missing values were imputed using the median of each variable.

6

Correct Data Types

The variables **age**, **employment_length_years**, **num_open_accounts**, and **months_since_oldest_account** had float data types, although they should be stored as integers. In addition, **application_date** had a string (object) data type instead of a date format.

The data types were converted to their appropriate formats.

7

Standardize Category Labels

It was also observed that the variables **home_ownership** and **loan_purpose** contained inconsistent label formatting.

The category labels were standardized to ensure consistency.

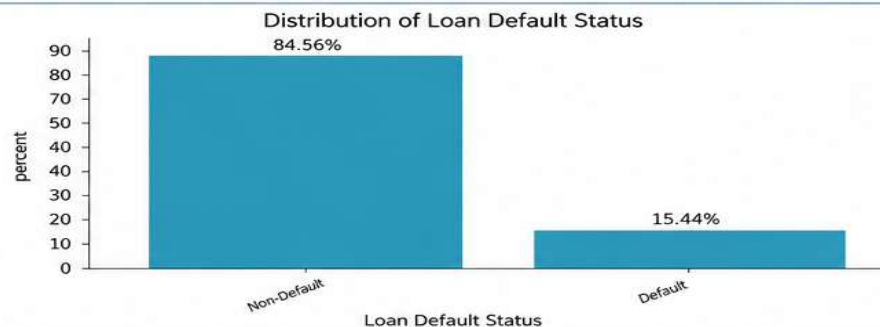
Exploratory Data Analysis (EDA)



1 Target Distribution



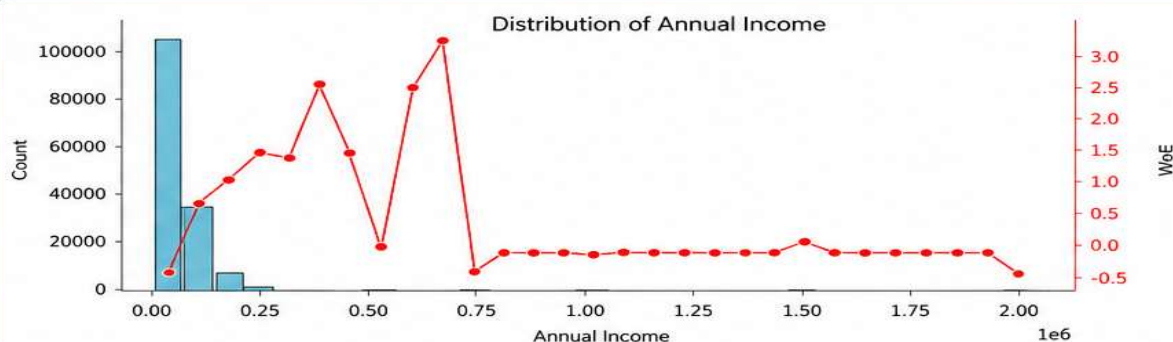
- From the target distribution diagram, we can see that people who defaulted make up 15.44% of the dataset, while non-defaults account for 84.56%.
- This shows that the dataset is imbalanced and further implies that there are many good customers and relatively few bad customers.



2 Annual Income



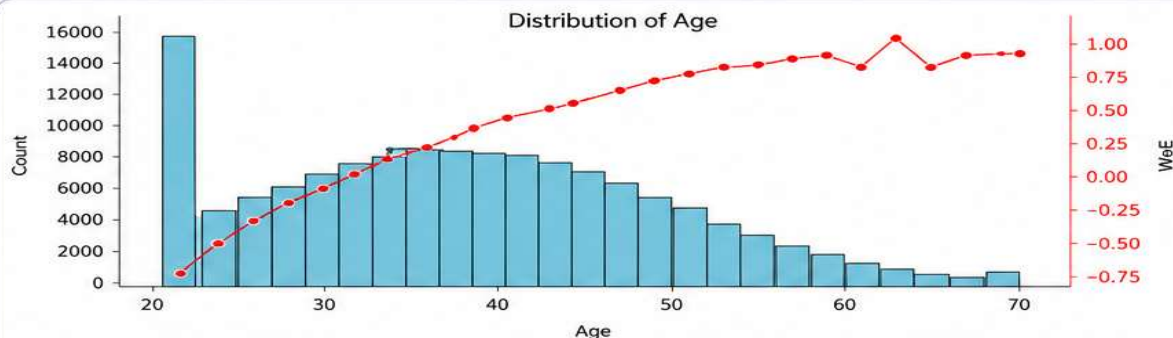
- The histogram shows many tall bars on the left side and fewer bars spread across the right side, indicating that annual income is right-skewed (has a long right tail) and may contain potential outliers.
- This suggests that most people earn lower incomes, while only a few people earn very high incomes.
- The WoE plot (red line) generally increases from one bin to another, indicating that the risk of default decreases as annual income increases.
- Although the relationship is not perfectly monotonic, this is likely because some bins at the higher income levels contain very few customers, causing the WoE values to be unstable in the later bins.



3 Age



- The histogram for age appears approximately normally distributed, with a larger concentration of younger applicants around age 21.
- The WoE plot (red line) is consistently increasing, showing the monotonic relationship that we expect.
- However, the curve increases at a decreasing rate, suggesting a non-linear relationship between age and the log-odds of default.
- The increasing WoE trend indicates that as age increases, the risk of default decreases. This suggests that younger applicants are generally more likely to default than older applicants.

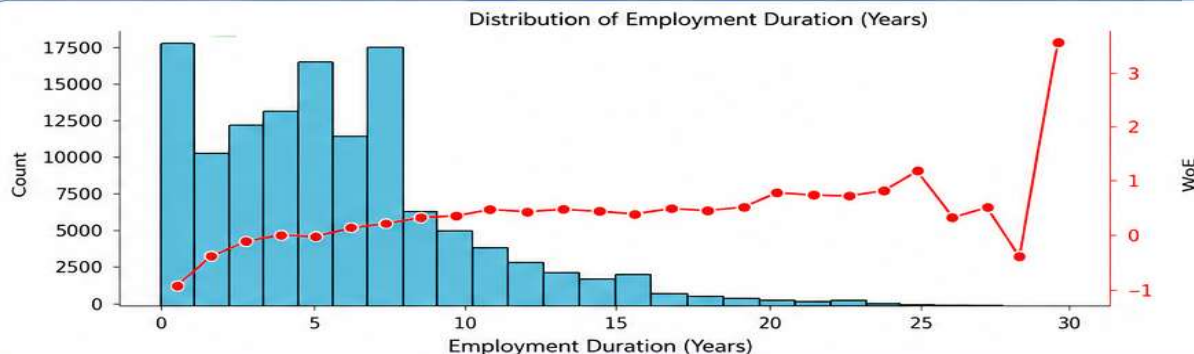


Exploratory Data Analysis (EDA)



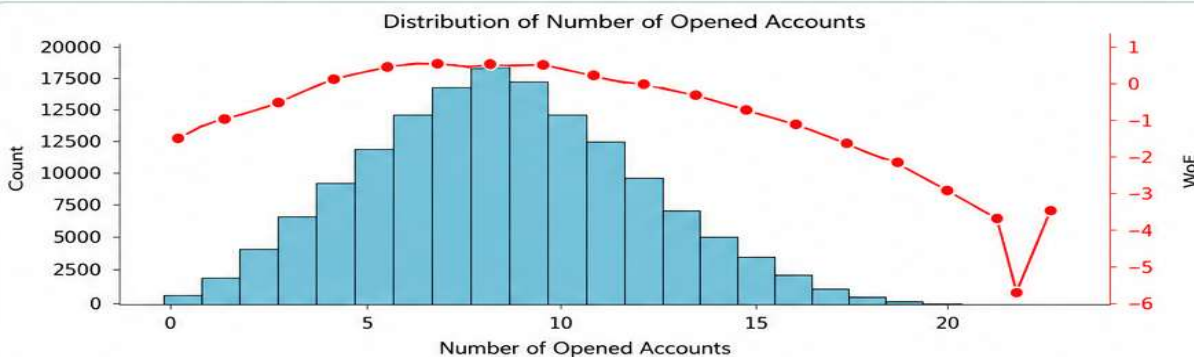
4 Employment Duration (Years)

- The histogram shows a slight right tail, with most people having employment durations between 0 and 10 years.
- The WoE plot (red line) generally increases, although it is not perfectly monotonic. The curved shape suggests that employment duration is not perfectly linearly related to the log-odds.
- The increasing trend implies that as employment duration increases, the risk of default decreases. This suggests that early-career professionals are generally more likely to default compared to more experienced individuals.



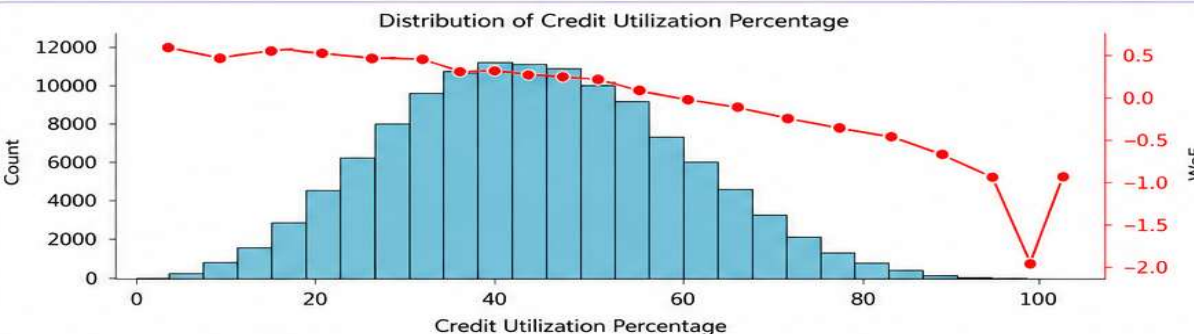
5 Number of Opened Accounts

- The histogram for the number of opened accounts appears approximately normally distributed, with most people having around 8 opened accounts.
- The WoE plot forms a quadratic-like shape, suggesting that the variable does not have a linear relationship with the log-odds, but rather a non-linear relationship.
- This indicates that when customers have very few accounts, opening additional accounts may initially reduce risk. However, after reaching approximately 8 accounts, the risk of default begins to increase again as more accounts are opened. This suggests that both very low and very high numbers of opened accounts may be associated with higher default risk, although having too many accounts appears riskier.



6 Credit Utilisation Percentage

- The distribution of credit utilisation percentage appears approximately normally distributed, with many customers using around 41% of their available credit.
- The WoE plot is consistently decreasing, showing a monotonic relationship, although it is not perfectly linear because the curve is slightly curved.
- The decreasing WoE trend indicates that as credit utilisation increases, the risk of default also increases. This implies that customers who use a smaller proportion of their available credit are generally less risky than those with high credit utilisation.



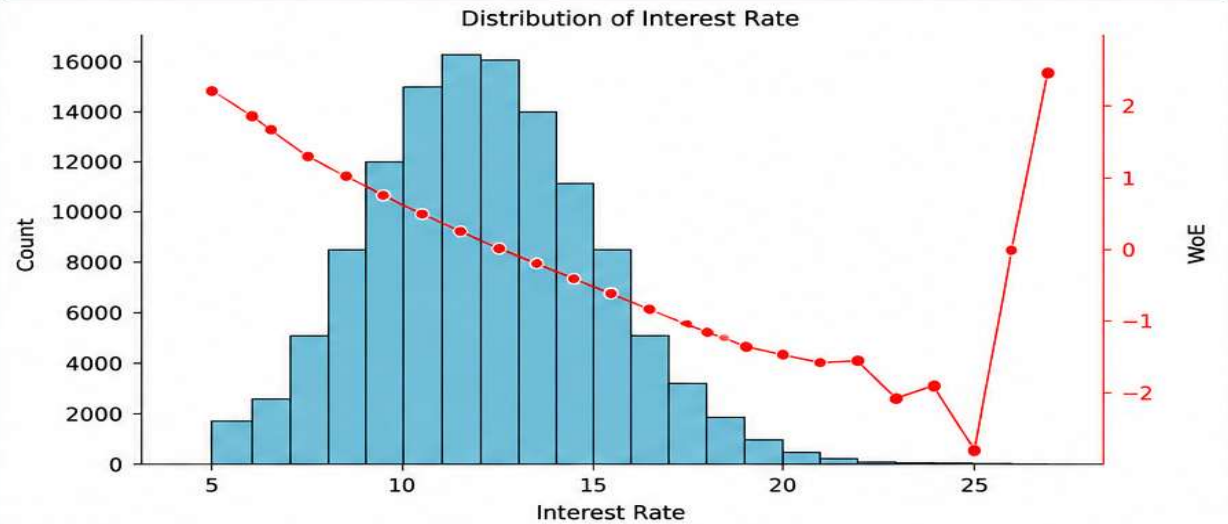
Exploratory Data Analysis (EDA)



7 Interest Rate



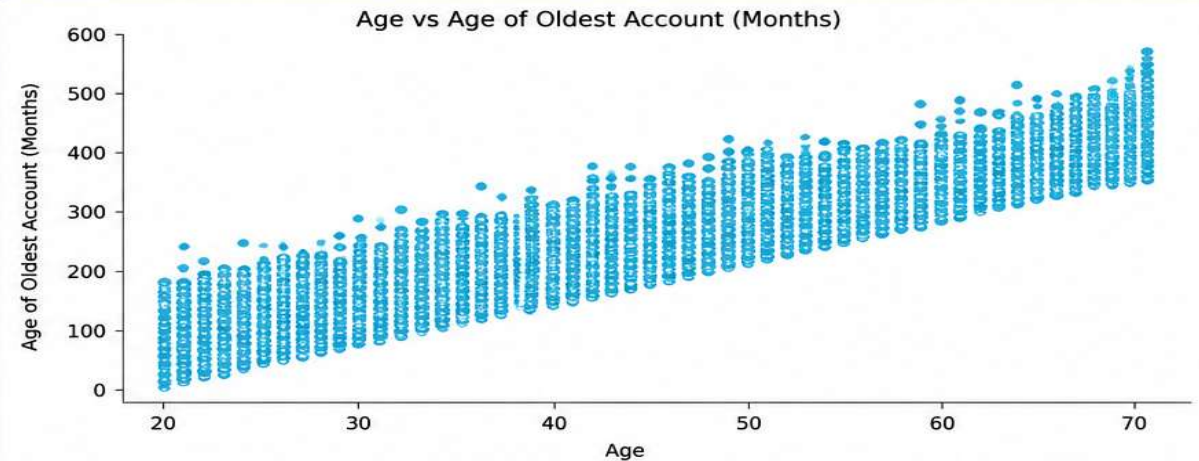
- The distribution appears approximately normal, with most customers being charged around a 12% interest rate.
- The WoE plot (red line) generally decreases, which is the monotonic relationship we expect, although there is some zig-zag behaviour at the extreme bins that may be due to potential outliers.
- This suggests that interest rate has a relatively linear relationship with the log-odds of default. We can further conclude that higher interest rates are associated with a higher risk of default. In other words, when loans become less affordable, borrowers are more likely to default.



8 Age vs Age of Oldest Account (Months)



- Age is positively and linearly related to months since the oldest account.
- This means that older individuals tend to have older credit accounts.
- The relationship suggests a strong correlation between the two variables.



Feature Engineering Choices



1

Transform **annual_income**

For feature engineering, we first consider **annual_income**. Since the variable was right-skewed and contained potential outliers, a new variable called **log_annual_income_capped** was created. First, the **annual_income** variable was capped to reduce the effect of extreme outliers, and then a logarithmic transformation was applied to the capped values. The logarithmic transformation helps improve the relationship between annual income and the log-odds by making the relationship more linear.



2

Apply **WoE Transformation to Key Features**

For the variables **age**, **home_ownership**, **num_open_accounts**, **credit_utilisation_pct**, and **employment_length_years**, the exploratory data analysis showed that these features had non-linear relationships with the log-odds of default. To better capture these relationships, new transformed variables ending with **_WoE** were created, for example **feature_WoE**.



3

Replace Values with **WoE**

For each variable, the values were separated into bins, and each observation was replaced with the corresponding Weight of Evidence (WoE) value for the bin to which it belonged. This transformation helps capture non-linear relationships while also making the variables more suitable for logistic regression.

Modeling Methodology



1 Logistic Regression as the Primary Model

For this credit risk model, logistic regression was selected as the primary modelling technique. The main reason for choosing logistic regression is that it is **highly interpretable** and suitable for **binary classification** problems, where the objective is to distinguish between good and bad customers by predicting the likelihood of default.

In addition, logistic regression produces **predicted probabilities**, which can be used to support lending decisions such as whether to approve or reject a loan application.

2 Train-Test Split

The dataset was split into training and testing sets using the variable named **set** in the data.

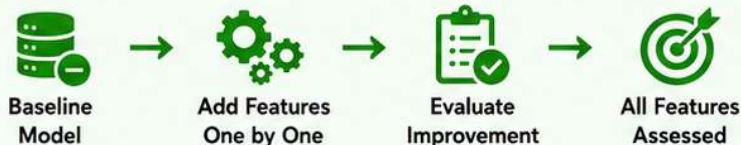
 **Training Set**
70%
of observations

 **Testing Set**
30%
of observations

3 Model Building Strategy

A **baseline** logistic regression model was first fitted after removing all forbidden features.

Thereafter, additional **engineered features** were added one at a time to evaluate whether they improved model performance. This process continued until all newly engineered features had been assessed.



4 Primary Objective: Improve AUC

The primary objective during model development was to improve the model's **AUC (Area Under the ROC Curve)**, since AUC measures how well the model distinguishes between default and non-default borrowers.

 **Better Discrimination**

 **Identify Risk More Accurately**

 **Better Lending Decisions**

Final Logistic Regression Model



Model Type

The final model was developed using logistic regression.



Objective

Predict the probability of default (p) for each borrower, where p is the probability of being a default (bad) customer.



Interpretation

The model estimates the log-odds of default. Positive coefficients increase the log-odds (higher default risk), while negative coefficients decrease the log-odds (lower default risk).



Variables

The model includes a mix of transformed (WoE and log) and original variables that were selected based on their predictive power.



Data Preparation

All variables used in the model were appropriately cleaned, engineered, and scaled where required.



Final Logistic Regression Equation

(All variables are scaled)

$$\log\left(\frac{p}{1-p}\right) = -0.4318$$

- -0.5754 (*log_annual_income_capped*)
- $+0.5387$ (*num_delinquencies_2yr*)
- -0.3895 (*age_woe*)
- -0.3275 (*num_open_accounts_woe*)
- $+0.2715$ (*num_hard_inquiries_6mo*)
- -0.2678 (*credit_utilisation_pct_woe*)
- -0.2371 (*employment_length_years_woe*)
- $+0.1012$ (*dti_ratio*)
- -0.0907 (*pct_accounts_current*)
- $+0.0835$ (*interest_rate*)
- -0.0430 (*home_WoE*)
- $+0.0262$ (*loan_amount*)

Interpretation of the Final Logistic Regression Model



Model Coefficients

Since the variables were scaled before fitting the logistic regression model, the coefficients represent the effect of a one standard deviation increase in each feature on the log-odds of default, holding all other variables constant.



Odds Ratios

The odds ratios describe the multiplicative change in the odds of default associated with a one standard deviation increase in a feature.



How to Read

- ↑ **Positive coefficient (OR > 1):** Increases the odds of default (higher risk)
- ↓ **Negative coefficient (OR < 1):** Decreases the odds of default (lower risk)



Key Takeaway

Positive coefficients increase default risk, while negative coefficients decrease default risk. The magnitude of the odds ratio shows the strength of the effect.

Feature	Coefficient	Odds Ratio (exp(coef))	Interpretation
↓ log_annual_income_capped	-0.5754	0.562	Negative coefficient and odds ratio < 1 indicate that higher income is associated with lower default risk. A one standard deviation increase decreases the odds of default by approximately 43.8%.
↑ num_delinquencies_2yr	0.5387	1.714	Positive coefficient and odds ratio > 1 indicate that more past delinquencies increase default risk. A one standard deviation increase increases the odds of default by approximately 71.4%.
↓ age_woe	-0.3895	0.678	Negative coefficient indicates that borrowers in lower-risk age groups are generally less likely to default.
↓ num_open_accounts_woe	-0.3275	0.721	Negative coefficient suggests that favourable account structures are associated with lower default risk.
↑ num_hard_inquiries_6mo	0.2715	1.312	Positive coefficient and odds ratio > 1 indicate that more recent hard inquiries are more likely to be associated with default.
↓ credit_utilisation_pct_woe	-0.2678	0.765	Negative coefficient shows that lower-risk credit utilisation groups are associated with lower default probabilities.
↓ employment_length_years_woe	-0.2371	0.789	Negative coefficient suggests that stronger employment profiles are associated with lower default risk.
↑ dti_ratio	0.1012	1.106	Positive coefficient indicates that higher debt-to-income ratios are associated with increased default risk.
↓ pct_accounts_current	-0.0907	0.913	Negative coefficient implies that borrowers with a higher proportion of current accounts are generally less risky.
↑ interest_rate	0.0835	1.087	Positive coefficient suggests that higher interest rates are associated with higher default risk.
↓ home_WoE	-0.0430	0.958	Small negative coefficient indicates a relatively weak relationship with default risk.
↑ loan_amount	0.0262	1.027	Positive coefficient suggests that larger loan amounts slightly increase default risk.

Performance Relative to Baseline



The current logistic regression model achieved an **AUC of 0.72**, outperforming the baseline logistic regression model (AUC = 0.68).



Absolute Improvement

+0.04

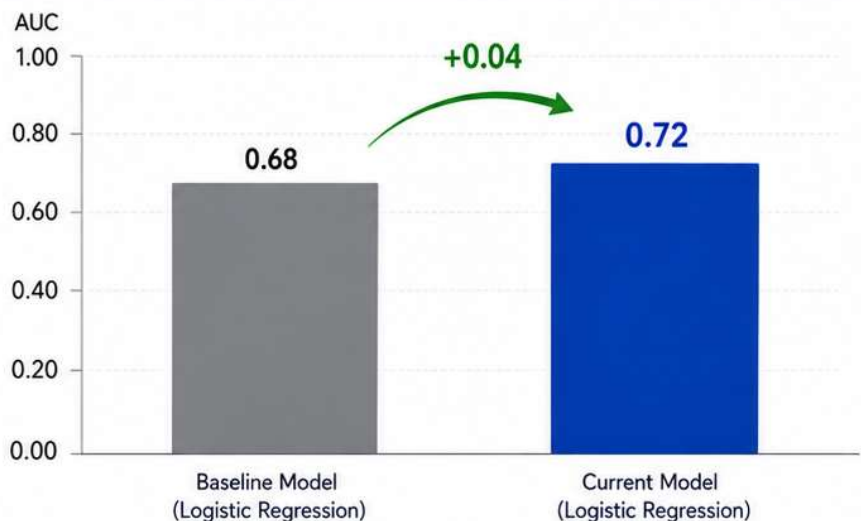


Relative Improvement

+5.9%

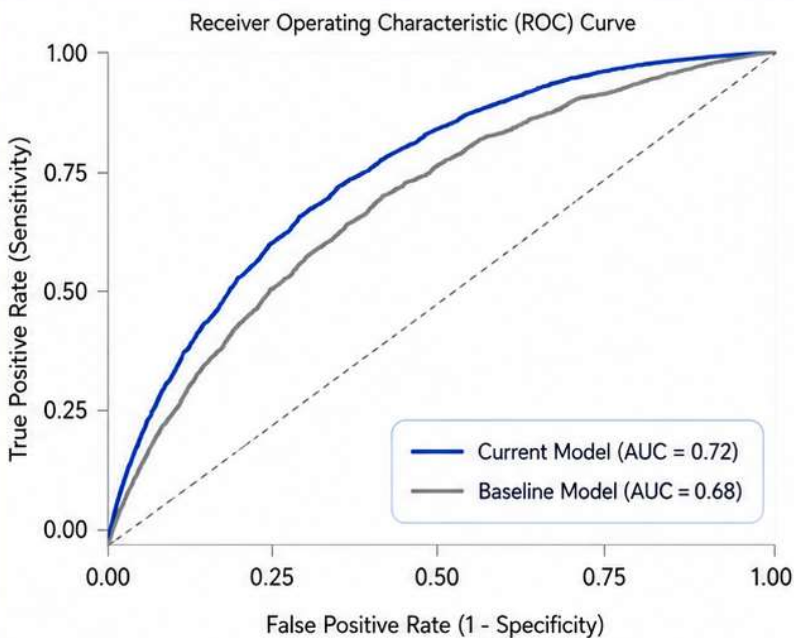
over the baseline

AUC Comparison



The current model demonstrates better discriminatory power, as measured by a higher AUC.

ROC Curve Comparison



Higher AUC indicates better ability to distinguish between defaulters and non-defaulters.